



EE712 - Embedded Systems Design

Electrical Engineering IIT Bombay

Laboratory Experiment - I

Problem set: 1.1

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Introduction to TIVA-C and CCS

- Perform all the steps mentioned from the Blink LED Example slide in “CCS Installation.pdf” to create a sample project. The Blink LED snippet code is available in the given link. LED_Blink_test
- After running the program, you must observe a blinking LED(Blue). Please make sure that you check what registers are changing when you execute a particular instruction in debug mode.
- Now, modify the code so the LED color cycles through Red, Green, and Blue.

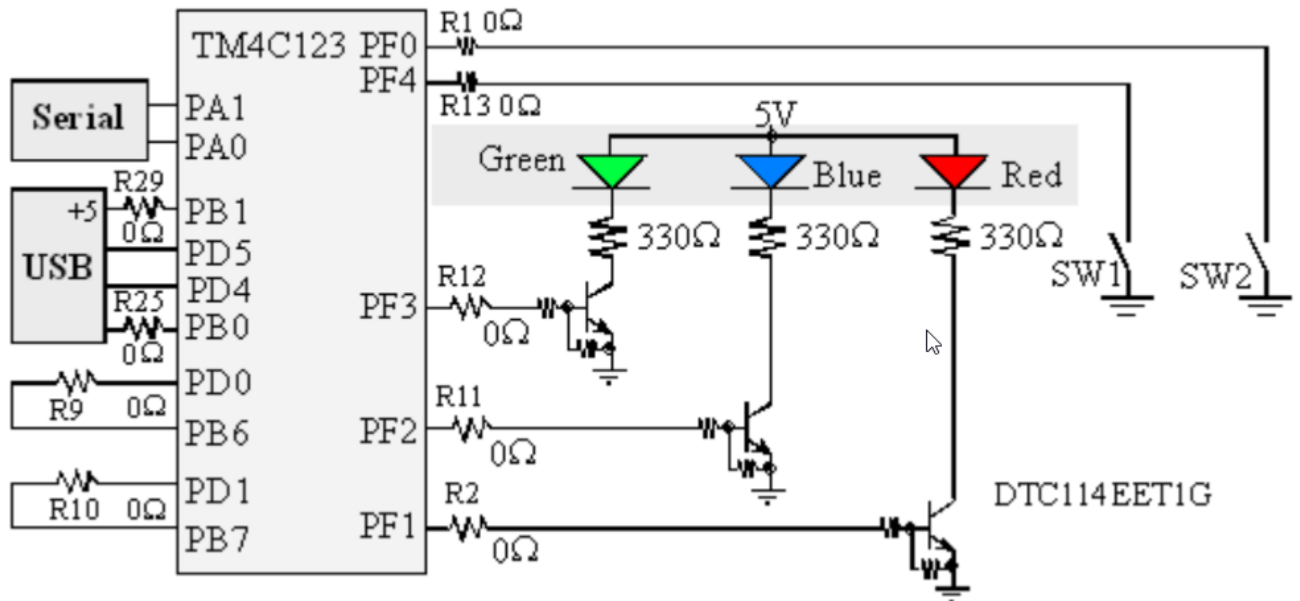


Figure 1: Connection of on-board switches and LEDs

Problem Statement

- Configure SW1 and SW2 such that Every time SW1 is pressed, toggle delay of LED should cycle through approximately 0.5s, 1s, 2s (Of any one color).
- Every time SW2 is pressed color of LED should cycle through Red, Green and Blue.
- The document TIVA-C userguide.pdf gives a brief overview of kit Tiva™ C Series TM4C123G LaunchPad Evaluation Board (EK-TM4C123GXL) which we are using for lab experiments.
- Please note that SW2 is connected to PF0 which is a special consideration GPIO pin (please refer TIVA-microcontroller-datasheet.pdf).

- Students can refer to **this** example on how to use GPIOs as interrupts.
- This pin is configured as a GPIO by default but is locked and can only be reprogrammed by unlocking the pin in the GPIO Lock (GPIOLOCK) register and dis-committing it by setting the GPIOCR (GPIO commit control) register.

Some useful tips to Solve the problem statement

Please follow the steps given below to unlock PF0 and to configure the port pins.

- Define MACROS globally before main for the base address of GPIOLOCK and GPIOCR.

```
# define LOCK_F (*((volatile unsigned long *)0x40025520))

# define CR_F (*((volatile unsigned long *)0x40025524))
```
- Write 0x4C4F434B to LOCK_F and GPIO_PIN_0 or 0x00000001 to CR_F after setting system clock and enabling peripherals (i.e. after SysCtlClockSet() and SysCtlPeripheralEnable()) in hardware_init function.

```
LOCK_F = 0x4C4F434BU;
CR_F = GPIO_PIN_0 | GPIO_PIN_4;
```

- Set the desired pins of PORT-7 as input and output pins.

```
GPIOPinTypeGPIOOutput(GPIO_PORTF_BASE, GPIO_PIN_1 | GPIO_PIN_2 | GPIO_PIN_3);
GPIOPinTypeGPIOInput(GPIO_PORTF_BASE, GPIO_PIN_0 | GPIO_PIN_4);
```
- Program the GPIO pad configuration to have weak pull-up.

```
GPIOPadConfigSet(GPIO_PORTF_BASE, GPIO_PIN_0 | GPIO_PIN_4,GPIO_STRENGTH_2MA,
GPIO_PIN_TYPE_STD_WPU);
```
- Please make the above-mentioned changes in the code given to you for lab 1 and proceed with the homework.